How execution-mode scores change same-challenge matching

A Same-challenge link recovered

Human · back_to_the_past

pwd → ls → cat → ls → cd → ls → cd → ls …

48 commands · detector Agent score 0.3683

Agent · back_to_the_past

curl → sed → curl → sed → which → nmap → sed → echo …

47 commands · detector Agent score 0.7221

Score

Behavior features only

0.5088 < 0.5131

No link · missed link

Behavior + mode scores

0.5482 ≥ 0.5127

Link · recovered

B Different-challenge link introduced

Human · back_to_the_past

curl → curl → curl → curl → mkdir → cd → ls → cat …

15 commands · detector Agent score 0.3411

Agent · glacier-exchange

curl → sed → nmap → curl → sed → mkdir → curl → sed …

72 commands · detector Agent score 0.7763

Score

Behavior features only

0.4738 < 0.5131

No link · correct rejection

Behavior + mode scores

0.5161 ≥ 0.5127

Link · spurious link

Each comparison: association score vs training-selected threshold. Command previews show the first 8 names; scoring uses full traces.
Two decisions from held-out public CTF tasks; same-challenge labels and fixed models. Aggregate effects are reported separately.